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SCLATO: SOFTWARE FOR ASCENT TRAJECTORY OPTIMISATION WITH APPLICATION TO CONVENTIONAL LAUNCHERS

A software product, SCLATO, has been developed to optimise the ascent trajectory of conventional launchers using a classic hybrid approach. The trajectory is parametrised by a reduced set of parameters. A subset, the tuneable parameters, is optimised by an optimisation package to maximise the payload mass while satisfying the mission constraints. SCLATO retains the capability of its precursor SARATO to optimise Ariane 5 ascent trajectories. Additionally, its configuration files can be customised to cover the user particular needs on launcher and mission. The underlying mathematical formulation is presented, as well as the application to the launch of a commercial launcher to LEO and a 2-stage Mars-to-orbit rocket. This paper is complemented by other paper presented to this conference and devoted to the generation of an initial guess of the trajectory by means of genetic algorithms (GA).

Un nouvel logiciel, SCLATO, optimise des trajectoires des lanceurs conventionnels avec une méthode hybride classique, où la trajectoire devient définie par un nombre reduit de paramètres, plusieurs desquels peuvent être adjustés au choix afin d'injecter une charge utile maximale dans l'orbite choisie, en même temps que les contraintes de mission sont respectées. SCLATO retient toutes les capacités de son prédécesseur SARATO pour optimiser les trajectoires du lanceur Ariane 5 et, en plus, il permet d'adapter ses fichiers de configuration à mesure des nouveaux lanceurs ou missions. Cet article fournit la formulation mathématique utilisée ainsi que l'application du logiciel au lancement (i) d'un lanceur commercial en orbite LEO et (ii) d'un vehicle Mars-orbite à deux étages. Un autre article, aussi présenté à cette conférence, se dévoue à la génération des valeurs initiales par moyen des algorithmes génétiques (GA).

1 – Introduction

GMV has developed a software product (SCLATO) to solve the problem of maximising the payload mass injected into a prescribed target orbit by a non-lifting, rocket-propelled launcher, while satisfying launch, in-flight and safety constraints.

SCLATO has evolved from SARATO, a software tool produced by GMV S.A. in the frame of a project led by SENER under a contract with CNES. SARATO was especially tailored for Ariane 5 ESC-B, the latest launcher of the Ariane 5 family. The cryogenic upper stage ESC-B mounts the new Vinci engine, with capability for multiple in-flight restarts. The restart feature enlarges the range of available target orbits. Apart from the classical launches into SSO, MEO or GTO orbits, or into escape trajectories, the ESC-B upper stage may perform perigee-raising manoeuvres at the first GTO apogee, in the case of a GTO+ launch, or may even perform direct injection into GEO. Multi-boost trajectories, in which separate payloads are injected into different orbits, are also feasible. In escape missions, it is possible to split the upper stage burn into two separate thrust arcs to get high declination asymptotes with little mass penalty. SARATO optimised all the above-mentioned trajectories, and was able to reproduce CNES-supplied design trajectories and performances[1].

From the original SARATO software and after extensive recoding, enhancements have been added to get to SCLATO, looking for higher flexibility in the definition of launchers and missions, as well as faster computation and robustness. It also shows growth potential. The resulting product is not only applicable to the Ariane 5 family but also to other conventional launchers in any scenario.

The paper poses the mathematical formulation of the dynamics and guidance, and shortly describes the method used to get an optimal trajectory. The paper also presents the application of SCLATO to two cases far away from Ariane 5 launches. The first case is the launch with a simplified model of Boeing's Delta III, derived from public sources[2,3]. The difference between the performances calculated with SCLATO and Boeing's official figures ranges from 10 to 70 kg in the reproduced missions. Such a good agreement proves that SCLATO is suitable for performance assessment as well as for generating trajectory data in the preliminary phases of mission analysis and design. The use of accurate, official data on aerodynamics, mass and propulsion will yield high quality results, but even public domain data on these areas are likely to be sufficient to get quick and useful estimates before formally contacting the marketing organisation for confirmation. The second application shown here is the ascent of a two-stage rocket from the Martian surface to a low orbit about Mars, where a return vehicle may be waiting.

2 – The Ascent Trajectory Problem

The problem of optimising an ascent trajectory can be translated into a boundary value problem with additional intermediate constraints. This problem is solved by a well-known hybrid optimisation method[4,5,6]. A general-purpose optimiser minimises a regular function that depends upon a reduced set of optimisation variables. This set fully characterises the mission: launcher (lift-off mass, upper stage fuel load), atmospheric gravity turn trajectory (launch azimuth and initial tilt angle), flight timeline (on/off times and separation times), and exo-atmospheric guidance law. In particular, the thrust steering satisfies in-flight optimality conditions established in Lawden[7] and Pontryagin[8] works, and its evolution is defined by the initial values of the state adjoints at the beginning of the exo-atmospheric thrust arcs. Therefore, the use of multiple coast arcs increases the number of optimisation variables. Also, every time an in-flight constraint is activated, the necessary conditions for optimality request the adjoints to experience a jump. This makes even harder to guess where the domain of convergence towards an optimal solution is.

2.1.Mathematical Formulation

2.1.1. Equations of Dynamics

In an inertial reference frame, the equations of motion are

$$\begin{aligned} \dot{\mathbf{r}} &= \mathbf{v} \\ \dot{\mathbf{v}} &= -\nabla U + (T/m)\,\mathbf{t} - (D/m\,v)\,\mathbf{v} \end{aligned} \tag{1}$$

where **r** is the position vector, **v** the velocity vector, U the gravitational potential, m the launcher mass, T the thrust, **t** the unit vector along the thrust direction, and D the aerodynamics drag.

The launcher mass is time-dependent and embodies the separate masses of the launcher elements

$$m(t) = \sum_{i=1}^{N(t)} \sum_{j=0}^{n_j} m_{i,j}(t) \tag{2}$$

where $N(t)$ is the number of elements remaining attached to the launcher at time t, and $m_{i,j}$ is the mass of the jth component of the ith element. These elements may be stages (active, burnout or yet to be ignited), payload, the fairing, a escape rocket or any item dragged by the launcher in its ascent. The number of elements decreases with time as the spent elements are discarded.

Some of these separate masses are constant (e.g. the inert masses of the stages). Others have a dynamics of their own (e.g. the propellants of a burning stage, the hydraulic fluid of the control, the boiling cryogenic propellants during a ballistic arc), and decrease at a constant rate or in a more complex manner depending upon thrust level and mixture ratio.

At a given time t, the launcher thrust considers only the contributions of the active (thrusting) elements,

$$T(t) = \sum_{i \in \{active\}} T_i \tag{3}$$

The thrust of an active element is usually decomposed in a constant term (thrust in vacuum) minus a deduction produced by the ambient pressure acting upon the set of exhaust nozzles of the element

$$T_i = T_{i,v} - p(h) A_{e,i} \tag{4}$$

where $T_{i,v}$ is the in-vacuo thrust, $p(h)$ the ambient pressure, altitude dependent, and $A_{e,i}$ the total exhaust area of the nozzles in the ith (active) element. The in-vacuo thrust may be set to a constant level or may experience changes with time. Even in the case of a rocket motor, other thrust models are possible, e.g., if the base pressure is taken into consideration at high altitude or if the motor mounts a plug nozzle. Obviously, in a ballistic (coast) arc, the thrust is null.

The aerodynamic drag D takes the simple form

$$D = \rho(h) V_{aero}{}^2 C_D(M) A_{aero} / 2 \tag{5}$$

where ρ is the atmospheric density, which depends upon the altitude h, V_{aero} the aerodynamic velocity (with respect to the moving atmosphere), C_D the drag coefficient (Mach dependent), and A_{aero} the reference area for aerodynamic actions. Both C_D and A_{aero} depend upon the launcher configuration, which evolves with time.

2.1.2. Guidance

The actual thrust direction **t** is the output of a guidance-control function, with its own dynamics

$$\dot{\mathbf{t}} = \mathbf{f}_c(\mathbf{t}_g, \mathbf{t}, t) \tag{6}$$

where $\mathbf{f_c}$ represents the fast dynamics of the control, which forces the actual thrust direction **t** to "pursuit" the desired thrust direction $\mathbf{t_g}$ generated by a guidance function.

During endo-atmospheric flight, transverse aerodynamic loads constraint the launcher to fly a zero-incidence trajectory (the gravity turn arc), and the guidance reduces to

$$\mathbf{t}_g = \frac{1}{\|\cdot\|} \mathbf{v}_{aero} \tag{7}$$

In exo-atmospheric flight, the thrust can be freely steered to any direction, and many guidance schemes can be devised. Only the ones currently implemented in SCLATO are described below.

<u>Optimal Implicit Guidance</u>

A necessary condition of optimality[4,7,8,10,12] is

$$\mathbf{t}_g = \frac{1}{\|\cdot\|} \boldsymbol{\lambda}_v \tag{8}$$

where $\boldsymbol{\lambda}_\mathbf{r}$ and $\boldsymbol{\lambda}_\mathbf{v}$ are the adjoint vectors to position and velocity, with dynamics governed by

$$\begin{aligned} \dot{\boldsymbol{\lambda}}_r &= (-\nabla^2 U) \cdot \boldsymbol{\lambda}_v \\ \dot{\boldsymbol{\lambda}}_v &= -\boldsymbol{\lambda}_r \end{aligned} \tag{9}$$

This implicit guidance scheme needs to integrate the adjoint dynamics in parallel with the original dynamics to compute the actual thrust direction. The guidance parameters are simply the initial values of the adjoint vectors. This guidance is truly optimal iif it satisfies the transversality condition at the final time, a condition that is not to be checked here.

The propagation of the adjoints may be alleviated in ballistic flight with the help of a useful property

$$\begin{Bmatrix} \boldsymbol{\lambda}_v(t_1) \\ -\boldsymbol{\lambda}_r(t_1) \end{Bmatrix} = \boldsymbol{\Phi}(t_1, t_0) \begin{Bmatrix} \boldsymbol{\lambda}_v(t_0) \\ -\boldsymbol{\lambda}_r(t_0) \end{Bmatrix} \tag{10}$$

where $\boldsymbol{\Phi}(t_1,t_0)$ is the transition matrix of the unperturbed motion from time t_0 to time t_1. Fortunately, analytical or semi-analytical solutions exist in several cases of interest (e.g., Keplerian motion and Keplerian motion perturbed by Earth oblateness).

<u>(Bi-)Linear Tangent Law</u>

The (bi-)linear tangent law is near-optimal guidance law* given by

$$\begin{aligned} \mathbf{t}_g &= \frac{1}{\|\cdot\|}\boldsymbol{\lambda}_v \\ \boldsymbol{\lambda}_v(t) &= -\boldsymbol{\lambda}_{r,0}(t - t_0) + \boldsymbol{\lambda}_{v,0} \end{aligned} \tag{11}$$

and the guidance parameters are the initial values of the adjoints.

* This guidance is optimal in presence of a uniform gravitational acceleration[7].

<u>Constant Thrust Direction</u>

This is the simplest guidance scheme, which turns out to be optimal in some simple scenarios**

$$\mathbf{t}_g = \mathbf{const.} \tag{12}$$

** Maximum range is attained with this guidance law in presence of a uniform gravitational acceleration[7].

<u>Open-Loop Profile of Thrust Direction</u>

In this guidance scheme, the thrust direction is a function of time prescribed by the user

$$\mathbf{t}_g = \mathbf{t}_g(t) \tag{13}$$

2.1.3. Control

Three types of control are considered

i) Perfect control: the actual thrust direction equals the commanded one.
ii) Constant rate: the thrust direction error decreases linearly with time as the actual thrust direction approaches the commanded one. This control is parameterised by the angular rate.
iii) Transfer function: the actual thrust direction is the output of a transfer function excited by the commanded thrust direction. This control is parameterised by the zeros and poles of the transfer function.

2.1.4. Environmental Models

The environment models define

i) Atmosphere properties: density, speed of sound, pressure and winds
ii) Gravitational field (gravitational constant, Jeffery constant J2, third body acceleration)
iii) Planet (or moon) shape
iv) Distance to the coast (Earth only)

as functions of position and, in some cases, time. The set of models implemented for Earth, Mars and Mercury can be easily expanded to include other bodies.

2.2.Trajectory Architecture

The ascent trajectory is decomposed in phases and the launcher in stages. A phase is characterised by

i) a parent phase, from which it inherits the initial values of position, velocity, masses, and guidance variables
ii) a launcher configuration, defined by the set of stages composing the launcher during the whole phase, and the aerodynamics properties for that configuration
iii) a set of equations of dynamics
iv) a scheme for guidance and control
v) a set of environmental models
vi) an event terminating the phase
vii) a set of numerical methods

And a launcher stage (including inert stages) is characterised by

i) inert mass
ii) consumable masses at lift-off
iii) propulsion data (not for inert stages): thrust profile in vacuum, nozzle exhaust area, mass rate profile for every consumable

An example of trajectory architectures is presented in Table 2.

2.3.Optimisation Method

2.3.1. Parametric Optimisation

Launcher and ascent trajectory are fully described by a reduced set of parameters: event times, masses, launch angles, and guidance and control parameters. In the optimal ascent trajectory the tuneable parameters should take those values that maximise the payload inserted into the prescribed orbit while satisfying the active constraints.

<u>Optimisation variables</u>

The vector of optimisation variables is comprised of the tuneable parameters. The number of optimisation variables increases with the use of multiple coast arcs and with the split of thrust arcs to account for the activation of in-flight constraints.

<u>Performance index</u>

The performance index to be maximised is the launcher final mass, which gives the last payload mass after deducing the upper stage inert mass. In a multi-boost launch, only the last payload is to be optimised.

<u>Constraints</u>

Two kinds of constraints are evaluated:

i) Algebraic constraints at the end of a phase, dependent upon the instantaneous state and time.
ii) Integral constraints over a phase, a quadrature of an algebraic function of state and time.

The set of constraints available for every phase is as follows

i) Orbital elements (classical and equinoctial)
ii) Hyperbolic excess velocity and impact parameters

iii) Instantaneous excess over aerothermodynamic thresholds (applies to dynamic pressure, heat rate and aerothermal flux)
iv) Integrated excess over aerothermodynamic thresholds
v) Splashdown location of burnout boosters and stages (Fig. 1).

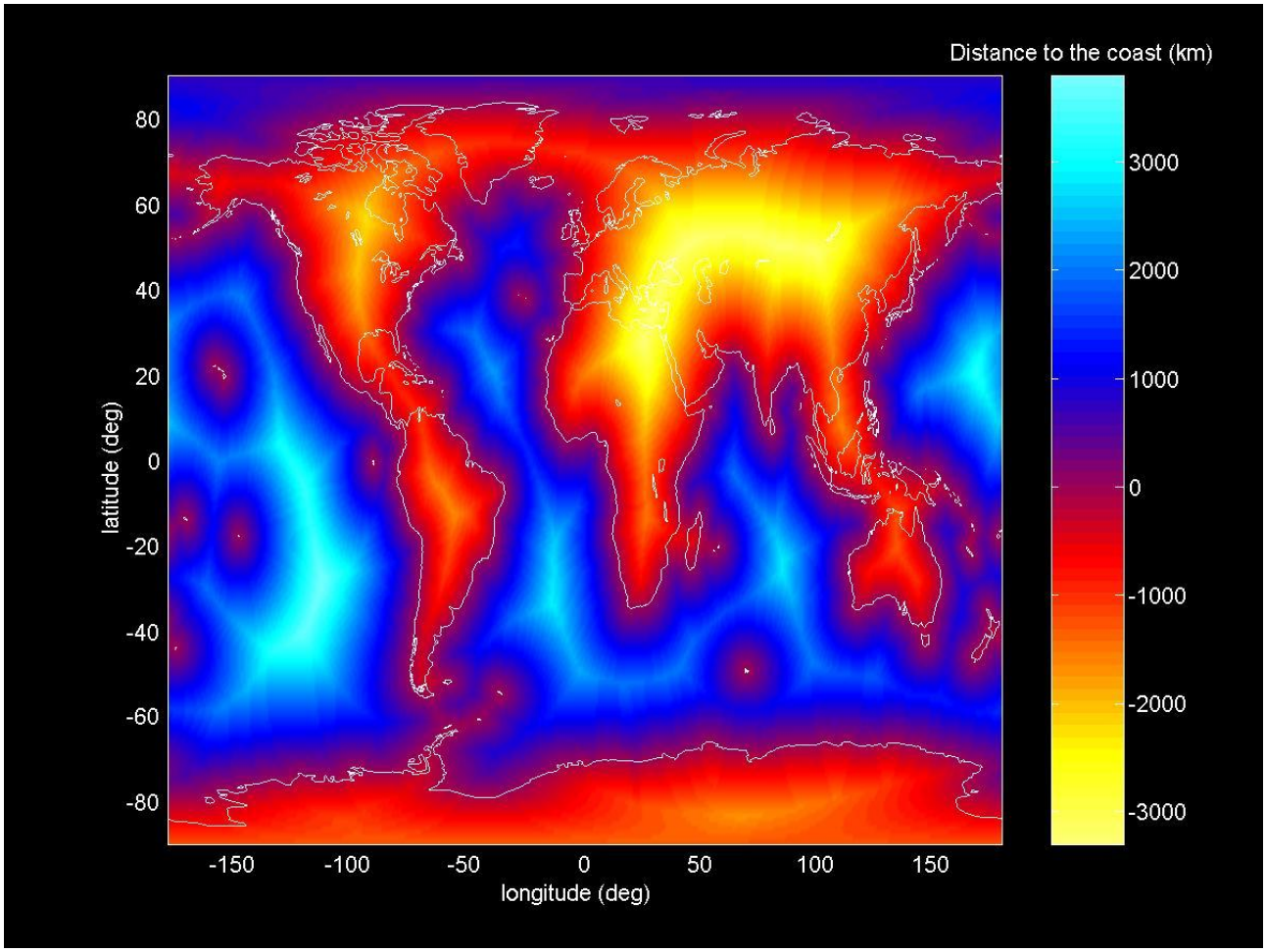


FIG. 1: Splashdown constraint: Map of distance to the closest shore.

The user can customise the set of applicable constraints by adequate choice in the above-mentioned set of available constraints. Similarly, any of them may be introduced as an equality or inequality one. This offers the capability to customise the constraint set to the particular needs of the user.

Optimisation method

The mathematical programming problem to be solved is to maximise the scalar performance index $f(\mathbf{x})$ of a vector argument $\mathbf{x}$, subject to a set of equality and inequality constraints, $\mathbf{g}_\mathbf{e}(\mathbf{x})$ and $\mathbf{g}_\mathbf{i}(\mathbf{x})$, respectively.

The solution is found with a penalty parameter technique[4,5,6,9,10,11]. A general-purpose optimisation package maximises an augmented performance index f_a that takes into consideration which constraints are violated

$$f_a = f - \frac{1}{\varepsilon} h\left(\begin{Bmatrix} \mathbf{g_e} \\ \mathbf{max(g_i, 0)} \end{Bmatrix}\right), \varepsilon > 0 \tag{14}$$

where ε is the penalty parameter and h is a penalty function satisfying

$$h(.) \geq 0\,,\ (h(\mathbf{y}) = 0 \Leftrightarrow \mathbf{y} = \mathbf{0}), h \xrightarrow[\|\mathbf{y}\| \to \infty]{} +\infty \tag{15}$$

An iterative scheme solves the problem for a given penalty parameter and then reduces it before restarting the search for a maximum. In the limit $\varepsilon \to 0$, the solution to maximising the augmented index becomes the solution to the original constrained problem.

The importance of the initial guess

The described optimisation method performs a local search. Continuation methods can be used to find the solution to a new problem by starting from the solution to a nearby problem. This approach is usually fruitful if such a nearby problem has been already solved, but does not guarantee that the found solution is the global maximum. Furthermore, the continuation manifold may end, and a global search may turn out to be mandatory. Similarly, the mission database of a new launcher starts from scratch, and an acceptable first guess will be appreciated since small gains in payload may pay off. Other paper[13] presented to this conference deals with a method to generate such an initial guess.

3 – A Few Applications

Two applications of the described software product SCLATO are presented here.

3.1.Conventional Launcher to LEO

A simplified model of Boeing's Delta III launcher has been produced from public sources[2,3]. This launcher is comprised of the elements listed in Table 1. Three missions from Cape Canaveral have been reproduced with a simplified model. Table 2 presents the corresponding (simplified) trajectory architecture for a launch to the International Space Station (ISS). Table 3 compares Boeing's official performances with the results produced with SCLATO. Minor differences appear, rather small ones given the scarcity of information. Fig.2 plots altitude, velocity, load factor and aerothermal flux profiles produced with SCLATO in a launch to ISS. The splash down constraint on the first stage impact point has been evaluated with the model mapped in Fig.1, which is a universal model depending only upon the Earth shoreline.

TAB. 1 : Boeing's Delta III launcher: Configuration.

Quantity	Item	Comments
3	GEM 46	Without thrust vectoring
3	GEM 46 TV	With thrust vectoring
3	GEM 46 AL	Air-lit version (enlarged nozzle)
1	core 1st stage	LOX-RP1
1	core 2nd stage	Cryogenic
1	Fairing	4-m Diameter
1	Payload	Depends upon the mission

TAB. 2 : Boeing's Delta III launcher: Simplified trajectory architecture for a launch to the ISS.

Phase / Parent	Active stages	Passive stages	Guidance/ Control	End event	Comments
1 / Lift-off	3 x GEM46 TV 3 x GEM46 Core 1st stage	3 x GEM46 AL Core 2nd stage Payload Fairing	Constant direction	Given time	Vertical rise
2/1			Constant rate	Given time	-
3/2			Constant direction	Zero incidence	-
4/3			Zero incidence	Given time	Ends at booster burn-out
5/4	Core 1st stage	3 x GEM46 TV 3 x GEM46 3 x GEM46 AL Core 2nd stage Payload Fairing	Constant direction	Given time	3 x GEM46 TV and 3 x GEM46 are burnout. The phase ends with the jettison of the burnout boosters.
6/5	3 x GEM46 AL Core 1st stage	Core 2nd stage Payload Fairing	Zero incidence / Transfer function	Given time	Phase ends with booster burn-out

7/6	Core 1st stage	Core 2nd stage Payload Fairing	Linear tangent / constant rate	Small control error	-
8/7			Linear tangent / Perfect	Aerothermal flux below threshold	The phase ends with an aerothermal flux low enough as to jettison the fairing
9/8		Core 2nd stage Payload		Given time	The phase ends when the core 1st stage is spent
10/9	-	Core 1st stage	None	Splash down	Ballistic flight of spent stage
11/9	-	Core 2nd stage Payload	Constant direction	Given time	A short coast arc to separate the core 1st stage
12/11	Core 2nd stage	Payload	Optimal / constant rate	Small control error	-
13/12			Optimal / perfect	Given time	Cut-off time is an optimisation variable

TAB. 3 : Boeing's Delta III launcher: Official performances vs. SCLATO results.

Mission	Payload mass	
	Boeing data	SCLATO results
LEO	8290 kg	8260 kg
Space Station	7300 kg	7310 kg
GTO	3810 kg	3740 kg

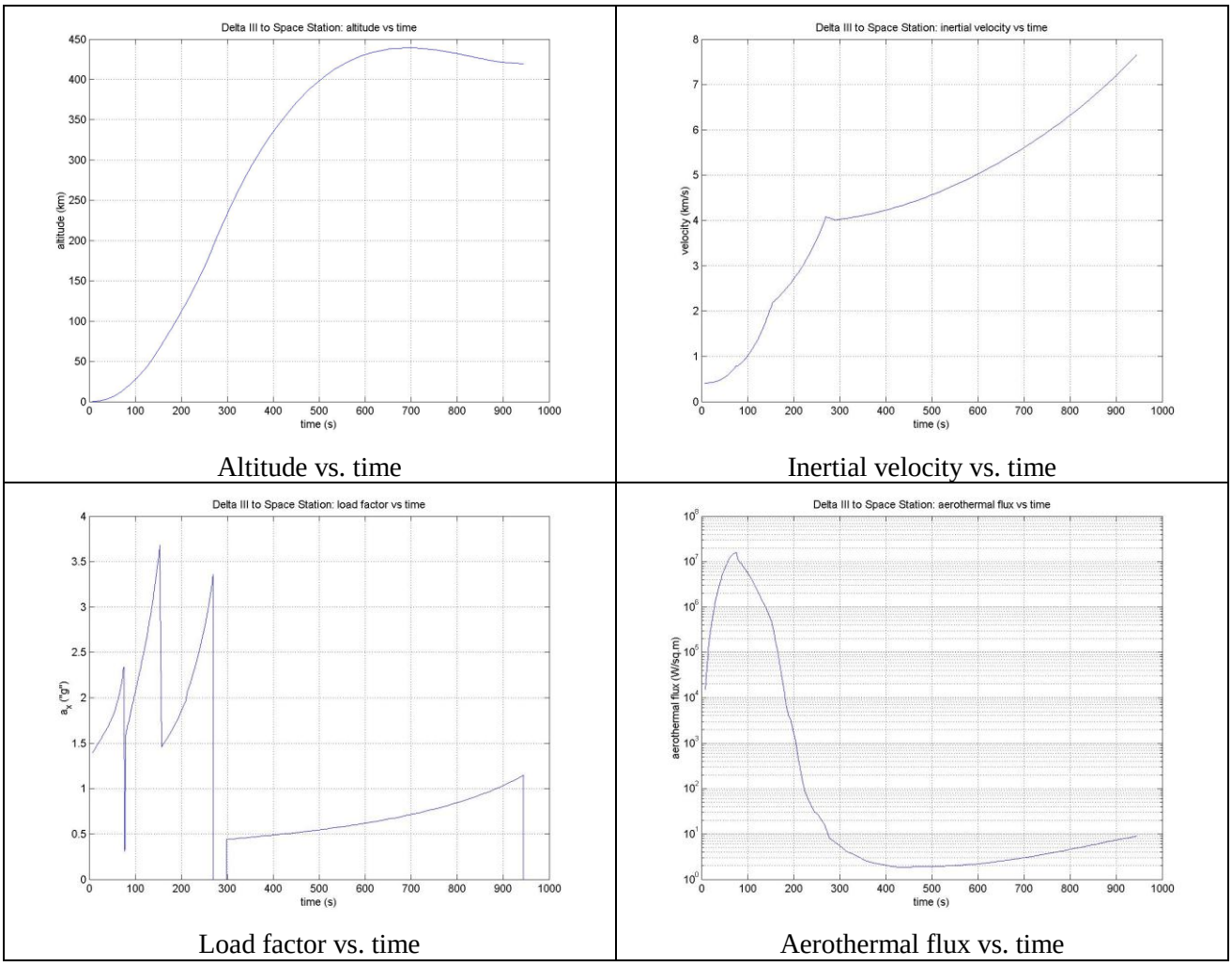


FIG. 2: Boeing's Delta III launcher: SCLATO plots for the launch to 407 km x 51.6 º (Space Station).

3.2.Mars to Orbit

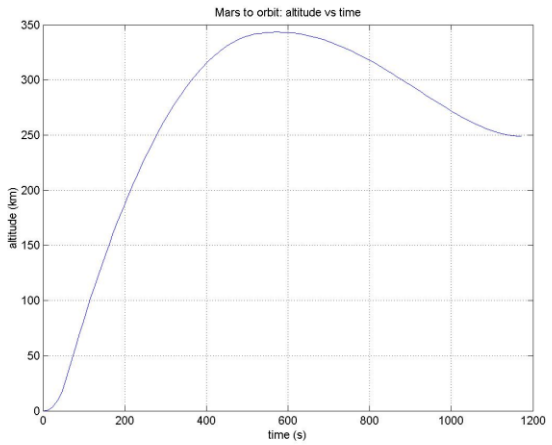
SCLATO has been applied also to the optimisation of the ascent trajectory of a hypothetical 2-stage Mars-to-orbit vehicle in a Mars sample return mission. Table 4 defines the vehicle and table 5 the rest of mission parameters. These values have been chosen for the sole purpose of testing SCLATO in this kind of mission, and do not represent a concept under study.

TAB. 4 : Mars to Orbit: Launcher description.

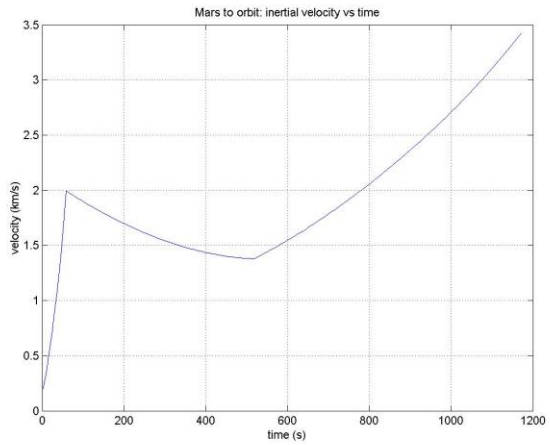
Parameter	Stage 1	Stage 2	Avionics+ Control	Payload
Inert mass (kg)	7.33	1.62	4.20	0.50
Propellant (kg)	21.98	6.48	-	-
Duration (s)	58.97	598.5	-	-
Thrust (N)	1041.8	32.4	-	-
Specific impulse (s)	285	305	-	-

TAB. 5 : Mars to Orbit: Mission description.

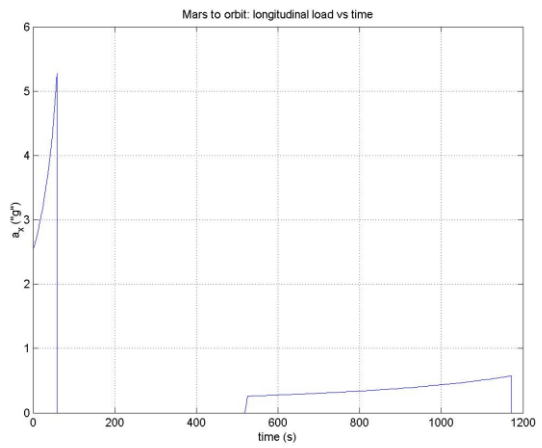
Launch latitude	40 º
Target orbit inclination	60 º
Target orbit altitude	250 km
1st Stage guidance	constant thrust direction
Coast arc	optimised duration
2nd Stage guidance	bi-linear tangent law



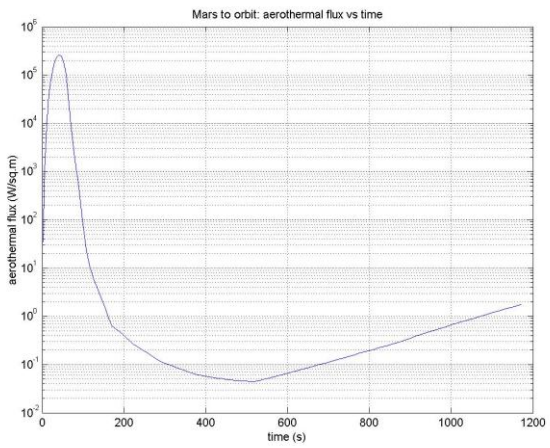
Altitude vs. time



Inertial velocity vs. time



Load factor vs. time



Aerothermal flux vs. time

FIG. 3: Mars to Orbit: SCLATO plots for the launch into a 250 km x 60 º orbit.

2 – Conclusions

SCLATO is a software product for ascent trajectory optimisation with application to conventional launchers. Additionally, it retains the capabilities of its precursor product, Sarato, to optimise Ariane 5 trajectories[1].

SCLATO has been successfully applied to the optimisation of launch trajectories from the Earth and Mars. It is easy to customise its configuration files to consider any conventional launcher from any planet.

The problem of obtaining an acceptable guess of the ascent trajectory is critical when a new launcher is considered or when the desired mission is too different from the ones already studied for that launcher. Other paper[13] presented to this conference deals with a method to generate such an initial guess. It also contains applications of SCLATO to other missions.

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